Dataset ls_hypercubes_<n>.cnf

This dataset can be reproduced with the python program ls_hypercubes.py:

```
>python ls_hypercubes.py <n>
```

where n is the size of the hypercube.
The dataset represents the number of n-dimensional hypercubes with vertices marked 0 or 1, such that no rectangle formed by four vertices has one diagonal 1-1 and the other diagonal 0-0. I mean all rectangles and not just the faces. The condition is necessary for a boolean function to be linearly separable. The question is whether it is also sufficient. I have proved that it is sufficient for n up to 7 by counting the hypercubes described above and verifying that the count is equal to OEIS A000609 - "Number of threshold functions of n or fewer variables". n=7 took 1 hour and 20 minutes with a direct count. n=6 took only 2 seconds. So I expect n=8 to be much harder, although I hope a #SAT solver can compute that case or even a higher one.
I have not tested the datasets with a #SAT, only with glucose SAT to get a solution.

For more information see my questions on stackexchange and other references:
https://en.m.wikipedia.org/wiki/Linear_separability
https://oeis.org/A000609
https://mathoverflow.net/q/462613/136218
https://math.stackexchange.com/q/4851044/573047

Dataset union_closed_<n>.cnf

This dataset can be reproduced with the python program union_closed_families.py:

```
>python union_closed_families.py <n>
```

where n is the number of elements of the universe of the union-closed family. The number of solutions is the number of union-closed families on at most n elements. A family of sets is said to be union-closed if the union of any two sets from the family belongs to the family. Union-closed families are very popular due to the union-closed sets conjecture (https://en.wikipedia.org/wiki/Union-closed_sets_conjecture) although here we just want to count them. The expected model count is at OEIS A102897. Cases n=8 and n=9 are not yet at OEIS, therefore they should be hard.
I have not tested the datasets with a #SAT, only with glucose SAT to get a solution.

For more information see:
https://oeis.org/A102897

License

All the material provided is licensed under the CC-BY license.

Reference

Fabio Visona – fabbio.visona@gmail.com